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COMPUTER NETWORKS (BACSE203)

CLASS ID: VL2026270101688 (B2+TB2 Slot) **FALL SEM 2026-27**

CASE STUDY ASSESSMENT

General Instructions to Students

Syllabus

Security Attacks, Trust and Threats, IPsec, Firewalls; SMTP, MIME, POP3; HTTP; FTP; Telnet; DNS, SNMP; and an introduction to SDN.

- Conduct online research before answering and cite every source with access date.
- Prefer RFC/IETF documents, official protocol/vendor documentation, research papers and credible technical sources.
- Include diagrams, packet captures, configuration evidence or screenshots wherever relevant.
- Answers should demonstrate analysis, design decisions and justification rather than only definitions.

Assessment Rubric

- Conceptual understanding and application – 20 marks
- Quality and relevance of research – 15 marks
- Security/technical analysis – 20 marks
- Architecture/design/packet-analysis evidence – 20 marks
- Justification and conclusions – 15 marks
- Presentation, diagrams and documentation – 10 marks

Total: 100 marks (Converted to 10 Marks)

Case Study 1
Insider Threat in a Financial Institution

Scenario
A bank suspects that an employee account is accessing systems outside normal responsibilities.

Research / Investigation Tasks
- Research trust models, insider threats, least privilege and segmentation.
- Study network indicators of suspicious access.

Questions
(a) Construct a threat model.

(b) Design controls limiting access to unrelated systems.

(c) Explain how firewall logs and network traffic support detection.

Case Study 2
IPsec for Secure Inter-Branch Communication

Scenario
A retail chain wants secure communication between 25 stores and its data center over the Internet.

Research / Investigation Tasks
- Research IPsec tunnel/transport modes and security services.
- Study site-to-site VPN architectures.

Questions
(a) Design an IPsec network for the stores and data center.

(b) Justify mode and security services.

(c) Analyse management challenges and propose a scalable architecture.

Case Study 3
Firewall Policy for a University Data Centre

Scenario
A data center hosts student portals, databases, DNS, mail and internal applications and has repeated unauthorized connection attempts.

Research / Investigation Tasks
- Study stateful firewalls, DMZs, segmentation and inbound/outbound filtering.
- Investigate logging and rule maintenance.

Questions
(a) Divide services into security zones.

(b) Create a firewall-policy matrix.

(c) Explain how permissive rules create attack paths and how rules should be reviewed.

Case Study 4
MIME and Secure Email Attachment Handling

Scenario
A company has malware incidents originating from email attachments.

Research / Investigation Tasks
- Study MIME content types and attachment representation.
- Research attachment threats, filtering and sandboxing.

Questions
(a) Explain how MIME carries attachments.

(b) Design an email gateway for screening dangerous attachments.

(c) Develop a policy for executable, compressed and unknown attachment types.

Case Study 5
POP3 Mailbox Access in a Low-Bandwidth Branch

Scenario
A remote branch has intermittent connectivity and users need email access.

Research / Investigation Tasks
- Research POP3 operation and mailbox retrieval.
- Study bandwidth, connectivity, authentication and security concerns.

Questions
(a) Explain POP3 and its limitations.

(b) Design an appropriate email-access workflow.

(c) Analyze security and reliability concerns of legacy mail access.

Case Study 6
Web Traffic Investigation After Credential Theft

Scenario
A company suspects users were tricked into submitting credentials through a fraudulent website.

Research / Investigation Tasks
- Study HTTP request/response communication and suspicious traffic indicators.
- Investigate limitations of unprotected HTTP and web-traffic controls.

Questions
(a) Describe how analysts can investigate suspicious HTTP traffic.

(b) Design network controls for malicious web destinations.

(c) Explain how DMZ placement can reduce risk to internal systems.

Case Study 7
FTP-Based Data Exchange for a Media Company

Scenario
A media company transfers large files between production offices using FTP.

Research / Investigation Tasks
- Study FTP control/data channels and firewall requirements.
- Research security limitations and network segmentation.

Questions
(a) Map FTP traffic through a firewall.

(b) Design a safer file-transfer architecture.

(c) Explain how firewall policies and monitoring can limit misuse.

Case Study 8
Legacy Telnet in an Industrial Control Network

Scenario
An industrial facility has legacy devices that only support Telnet and cannot immediately replace them.

Research / Investigation Tasks
- Study Telnet plaintext communication and compensating controls.
- Research segmentation, firewall restrictions and jump-host concepts.

Questions
(a) Design an isolated architecture for Telnet-enabled devices.

(b) Develop firewall restrictions for administrative access.

(c) Explain residual risks and a long-term migration strategy.

Case Study 9
DNS Security and Availability for an Online Service

Scenario
An online service has intermittent DNS failures and suspects malicious manipulation.

Research / Investigation Tasks
- Research DNS hierarchy, caching and record types.
- Study DNS spoofing/cache poisoning, redundancy and authoritative-server protection.

Questions
(a) Develop a diagnostic procedure.

(b) Design a resilient DNS architecture.

(c) Explain how malicious DNS information can redirect users and discuss countermeasures.

Case Study 10
SNMP-Based Predictive Network Monitoring

Scenario
A university wants to identify overloaded interfaces and failing devices before users are affected.

Research / Investigation Tasks
- Research SNMP managers, agents, management information and secure deployment.
- Study thresholds, alerts and historical trends.

Questions
(a) Design a monitoring-dashboard specification.

(b) Define threshold alerts for five network conditions.

(c) Explain how historical SNMP data can support failure prediction and capacity planning.

Case Study 11
Secure Public-Facing Services for an Online Bank

Scenario
An online bank exposes web, DNS and email services while protecting transaction-processing systems.

Research / Investigation Tasks
- Study DMZ architecture, firewall rules and attack paths from public servers.
- Research segmentation and controlled outbound communication.

Questions
(a) Design Internet-facing DMZ and internal security zones.

(b) Create a high-level firewall rule matrix.

(c) Analyze compromise of a public web server and explain how segmentation limits damage.

Case Study 12
SDN-Based Micro-Segmentation for a Smart Campus

Scenario
A university has student, faculty, IoT, laboratory and guest networks. Conventional VLAN/firewall changes are slow, so SDN is being considered.

Research / Investigation Tasks
- Research SDN architecture, controller functions and control/data-plane separation.
- Study SDN-based segmentation and security-policy enforcement.

Questions
(a) Design an SDN architecture separating the five user/device groups.

(b) Develop example controller policies between the groups.

(c) Evaluate SDN-based security-policy management against conventional distributed configuration.